

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1473

South Neal 100 kV
April 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1473. The Interconnection Customer proposes a 198.5 MW natural gas combined cycle facility interconnecting at the South Neal 100 kV substation in the Ohio area. The request was received on May 05, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 379 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,585,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Wilmington Station C Tap / S Neal 100 KV
Contingency: Hintz / Sheldon Springs Hydro Electric 100 KV
Loading: 152.1% (Rating: 145 MVA)
DFAX: 36.2%
Pre-project loading: 102.5%
2. Facility: Downs / Alps 100 KV
Contingency: A Whoch Substation / Smith Electrical 100 KV
Loading: 142.5% (Rating: 129 MVA)
DFAX: 15.4%
Pre-project loading: 118.7%
3. Facility: Wilmington Station C Tap / Downs 100 KV
Contingency: Smith Electrical / A Whoch 100 KV
Loading: 133.4% (Rating: 158 MVA)
DFAX: 15.4%
Pre-project loading: 114.0%
4. Facility: Mariposa / Acec Brooks 100 KV
Contingency: Wilmington Station C Tap / Downs 100 KV
Loading: 118.2% (Rating: 150 MVA)
DFAX: 6.1%
Pre-project loading: 110.2%
5. Facility: Southwind / Wilmington Station C Tap 100 KV
Contingency: Wilmington Sta C Tap / S Neal 100 KV Line
Loading: 115.0% (Rating: 124 MVA)
DFAX: 7.3%
Pre-project loading: 103.4%

6. Facility: Polypro / Carr Draw 100 KV
 Contingency: Breeze / Barrett 230 KV line
 Loading: 114.2% (Rating: 134 MVA)
 DFAX: 9.2%
 Pre-project loading: 100.5%
7. Facility: Mosher / Maripsa 100 KV
 Contingency: Wilmington Sta C Tap / Downs Substation 100 KV line
 Loading: 111.1% (Rating: 140 MVA)
 DFAX: 6.1%
 Pre-project loading: 102.6%
8. Facility: Hintz / Sheldon Springs Hydro Electric 100 KV Line
 Contingency: Wilmington Station C Tap / S Neal 100 KV line
 Loading: 107.3% (Rating: 146 MVA)
 DFAX: 14.1%
 Pre-project loading: 88.1%
9. Facility: Tilcon / South Neal 100 KV line
 Contingency: Wilmington Station C Tap / S Neal 100 KV
 Loading: 103.3% (Rating: 168 MVA)
 DFAX: 100.0%
 Pre-project loading: 14.7%
10. Facility: Sheldon Springs Hydro Electric / Kleburg 100 KV line
 Contingency: Wilmington Sta C Tap / S Neal 100 KV
 Loading: 103.1% (Rating: 138 MVA)
 DFAX: 14.1%
 Pre-project loading: 82.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Neal / Wilmington Station C Tap Substation 100 KV Line	Rebuild line with higher-capacity conductor	\$3,380,000	100% (\$3,380,000)
Downs / Alps 100 KV	Rebuild line with higher-capacity conductor	\$1,684,000	100% (\$1,684,000)
Wilmington Sta C Tap / Downs 100 KV	Rebuild line with higher-capacity conductor	\$943,000	100% (\$943,000)
Acec Brooks / Maiposa 100 KV	Rebuild line with higher-capacity conductor	\$3,694,000	100% (\$3,694,000)
Southwind / Wilmington Station C Tap 100 KV	Rebuild line with higher-capacity conductor	\$1,349,000	100% (\$1,349,000)
Carr Draw / Polypro 100 KV	Rebuild line with higher-capacity conductor	\$1,263,000	48% (\$610,000)
Mariposa / Mosher 100 KV	Rebuild line with higher-capacity conductor	\$2,086,000	100% (\$2,086,000)
Sheldon Springs Hydro Electric / Hintz 100 KV	Reconductor line	\$430,000	100% (\$430,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tilcon / S Neal 100 KV	Reconductor line	\$353,000	100% (\$353,000)
Kleburg Substation / Sheldon Springs Hydro Electric 100 KV	Reconductor line	\$1,056,000	100% (\$1,056,000)
	Total allocated to project		\$15,585,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.